



Derivative Formulas for JEE, Class 12th, and Class 11th

Differentiation is the cornerstone of calculus and forms one of the most critical topics in JEE Main and Advanced mathematics syllabus. This comprehensive guide covers all essential derivative formulas, rules, and techniques required for mastering calculus problems in competitive examinations and board exams.

JEE Derivative Formulas Reference

Category	Function	Derivative
Basic Functions	$f(x) = c$	$f'(x) = 0$
	$f(x) = x$	$f'(x) = 1$
	$f(x) = x^n$	$f'(x) = nx^{n-1}$
	$f(x) = \sqrt{x}$	$f'(x) = 1/(2\sqrt{x})$
	$f(x) = 1/x$	$f'(x) = -1/x^2$
Exponential/Log	$f(x) = e^x$	$f'(x) = e^x$
	$f(x) = a^x$	$f'(x) = a^x \ln(a)$
	$f(x) = \ln(x)$	$f'(x) = 1/x$
	$f(x) = \log_a(x)$	$f'(x) = 1/(x \ln(a))$
Trigonometric	$f(x) = \sin(x)$	$f'(x) = \cos(x)$
	$f(x) = \cos(x)$	$f'(x) = -\sin(x)$
	$f(x) = \tan(x)$	$f'(x) = \sec^2(x)$
	$f(x) = \cot(x)$	$f'(x) = -\csc^2(x)$
	$f(x) = \sec(x)$	$f'(x) = \sec(x)\tan(x)$

Essential Derivative Formulas for JEE, Class 11th & 12th

Basic Derivative Formulas

Fundamental Derivatives

The foundation of differentiation begins with understanding these basic formulas that are derived from the first principle of limits: [\[1\]](#) [\[2\]](#) [\[3\]](#)

Elementary Functions:

- $\frac{d}{dx}(c) = 0$ (where c is a constant)

- $\frac{d}{dx}(x) = 1$
- $\frac{d}{dx}(x^n) = nx^{n-1}$ (Power Rule)
- $\frac{d}{dx}\left(\frac{1}{x}\right) = -\frac{1}{x^2}$
- $\frac{d}{dx}(\sqrt{x}) = \frac{1}{2\sqrt{x}}$

Generalized Power Rule:

For any real number n : $\frac{d}{dx}(x^n) = nx^{n-1}$

This includes:

- $\frac{d}{dx}(x^{-n}) = -nx^{-n-1}$
- $\frac{d}{dx}(x^{1/n}) = \frac{1}{n}x^{(1/n)-1}$

First Principle of Derivatives

The derivative is defined using the limit: ^[2] ^[4]

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

This fundamental definition forms the basis for deriving all other differentiation formulas and is essential for understanding the concept of instantaneous rate of change.

DERIVATIVE RULES CHART

GENERAL DERIVATIVE RULES

Constant Rule	$\frac{d}{dx}[c] = 0$
Constant Multiple Rule	$\frac{d}{dx}[cf(x)] = cf'(x)$
Sum Rule	$\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x)$
Difference Rule	$\frac{d}{dx}[f(x) - g(x)] = f'(x) - g'(x)$
Product Rule	$\frac{d}{dx}[f(x) \cdot g(x)] = f'(x)g(x) + f(x)g'(x)$
Quotient Rule	$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}$
Chain Rule	$\frac{d}{dx}[f(g(x))] = f'(g(x))g'(x)$

DERIVATIVE RULES FOR PARTICULAR FUNCTIONS

FUNCTION	BASIC RULE	CHAIN RULE
Power	$\frac{d}{dx}[x^n] = nx^{n-1}$	$\frac{d}{dx}[u^n] = nu^{n-1} \cdot u'$
TRIGONOMETRIC FUNCTIONS		
Sine	$\frac{d}{dx}[\sin(x)] = \cos(x)$	$\frac{d}{dx}[\sin(u)] = \cos(u) \cdot u'$
Cosine	$\frac{d}{dx}[\cos(x)] = -\sin(x)$	$\frac{d}{dx}[\cos(u)] = -\sin(u) \cdot u'$
Tangent	$\frac{d}{dx}[\tan(x)] = \sec^2(x)$	$\frac{d}{dx}[\tan(u)] = \sec^2(u) \cdot u'$
Cosecant	$\frac{d}{dx}[\csc(x)] = -\csc(x)\cot(x)$	$\frac{d}{dx}[\csc(u)] = -\csc(u)\cot(u) \cdot u'$
Secant	$\frac{d}{dx}[\sec(x)] = \sec(x)\tan(x)$	$\frac{d}{dx}[\sec(u)] = \sec(u)\tan(u) \cdot u'$
Cotangent	$\frac{d}{dx}[\cot(x)] = -\csc^2(x)$	$\frac{d}{dx}[\cot(u)] = -\csc^2(u) \cdot u'$
INVERSE TRIGONOMETRIC FUNCTIONS		
Arcsine	$\frac{d}{dx}[\sin^{-1}(x)] = \frac{1}{\sqrt{1-x^2}}$	$\frac{d}{dx}[\sin^{-1}(u)] = \frac{1}{\sqrt{1-u^2}} \cdot u'$
Arccosine	$\frac{d}{dx}[\cos^{-1}(x)] = \frac{-1}{\sqrt{1-x^2}}$	$\frac{d}{dx}[\cos^{-1}(u)] = \frac{-1}{\sqrt{1-u^2}} \cdot u'$
Arctangent	$\frac{d}{dx}[\tan^{-1}(x)] = \frac{1}{1+x^2}$	$\frac{d}{dx}[\tan^{-1}(u)] = \frac{1}{1+u^2} \cdot u'$
Arccosecant	$\frac{d}{dx}[\csc^{-1}(x)] = \frac{-1}{ x \sqrt{x^2-1}}$	$\frac{d}{dx}[\csc^{-1}(u)] = \frac{-1}{ u \sqrt{u^2-1}} \cdot u'$
Arcsecant	$\frac{d}{dx}[\sec^{-1}(x)] = \frac{1}{ x \sqrt{x^2-1}}$	$\frac{d}{dx}[\sec^{-1}(u)] = \frac{1}{ u \sqrt{u^2-1}} \cdot u'$
Arccotangent	$\frac{d}{dx}[\cot^{-1}(x)] = \frac{-1}{1+x^2}$	$\frac{d}{dx}[\cot^{-1}(u)] = \frac{-1}{1+u^2} \cdot u'$
EXPONENTIAL FUNCTIONS		
Exponential (base e)	$\frac{d}{dx}[e^x] = e^x$	$\frac{d}{dx}[e^u] = e^u \cdot u'$
Exponential (base a)	$\frac{d}{dx}[a^x] = a^x \ln(a)$	$\frac{d}{dx}[a^u] = a^u \ln(a) \cdot u'$
LOGARITHMIC FUNCTIONS		
Natural Logarithm	$\frac{d}{dx}[\ln(x)] = \frac{1}{x}$	$\frac{d}{dx}[\ln(u)] = \frac{1}{u} \cdot u'$
Logarithm (base a)	$\frac{d}{dx}[\log_a(x)] = \frac{1}{x \cdot \ln(a)}$	$\frac{d}{dx}[\log_a(u)] = \frac{1}{u \cdot \ln(a)} \cdot u'$

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Derivative rules chart covering general rules and formulas for trigonometric, inverse trigonometric, exponential, and logarithmic functions with chain rule examples.

Differentiation Rules

Differentiation Rules for JEE

Rule	Formula	When to Use	Example
Constant Rule	$\frac{d}{dx}(c) = 0$	When differenti...	$\frac{d}{dx}(5) = 0$
Power Rule	$\frac{d}{dx}(x^n) = nx^{n-1}$	For polynomial ...	$\frac{d}{dx}(x^3) = 3x^2$
Constant Multip	$\frac{d}{dx}(cf(x)) = c \cdot f'(x)$	When constant m...	$\frac{d}{dx}(3x^2) = 3 \cdot 2x = 6x$
Sum/Difference	$\frac{d}{dx}(f \pm g) = f' \pm g'$	For sums/differ...	$\frac{d}{dx}(x^2 + \sin x) = 2x + \cos x$
Product Rule	$\frac{d}{dx}(uv) = u'v + uv'$	For products of...	$\frac{d}{dx}(x \cdot \sin x) = 1 \cdot \sin x + x \cdot \cos x$
Quotient Rule	$\frac{d}{dx}(u/v) = (u'v - uv')/v^2$	For quotients o...	$\frac{d}{dx}(\sin x/x) = (x \cos x - \sin x)/x^2$
Chain Rule	$\frac{d}{dx}(f(g(x))) = f'(g(x)) \cdot g'(x)$	For composite f...	$\frac{d}{dx}(\sin(x^2)) = \cos(x^2) \cdot 2x$
Logarithmic Dif	If $y = f(x)$, then $y'/y = (\ln f(x))'$	For complex pro...	$y = x^x$, $\ln y = x \ln x$, $y'/y = \ln x + 1$
Implicit Differ	Differentiate both sides w.r.t. x	When y is not i...	$x^2 + y^2 = 1 \rightarrow 2x + 2y \cdot dy/dx = 0$
Parametric Diff	$dy/dx = (dy/dt)/(dx/dt)$	When x, y given ...	$x = t^2$, $y = t^3 \rightarrow dy/dx = 3t^2/2t = 3t/2$

Differentiation Rules and Techniques for JEE Preparation

Basic Rules of Differentiation

Constant Rule:

$$\frac{d}{dx}(c) = 0$$

Constant Multiple Rule:

$$\frac{d}{dx}[cf(x)] = c \cdot f'(x)$$

Sum and Difference Rules:

$$\frac{d}{dx}[f(x) \pm g(x)] = f'(x) \pm g'(x)$$

Product Rule

For the product of two functions $u(x)$ and $v(x)$:^[1] ^[5] ^[6]

$$\frac{d}{dx}[u(x) \cdot v(x)] = u'(x) \cdot v(x) + u(x) \cdot v'(x)$$

Extended Product Rule:

For three functions: $\frac{d}{dx}[uvw] = u'vw + uv'w + uvw'$

Memory Aid: "First derivative times second as is, plus first as is times second derivative"

Quotient Rule

For the quotient of two functions:^{[1] [5] [6]}

$$\frac{d}{dx} \left[\frac{u(x)}{v(x)} \right] = \frac{v(x) \cdot u'(x) - u(x) \cdot v'(x)}{[v(x)]^2}$$

Memory Aid: "Low d-high minus high d-low, over low squared"

Chain Rule

For composite functions $f(g(x))$:^{[6] [7]}

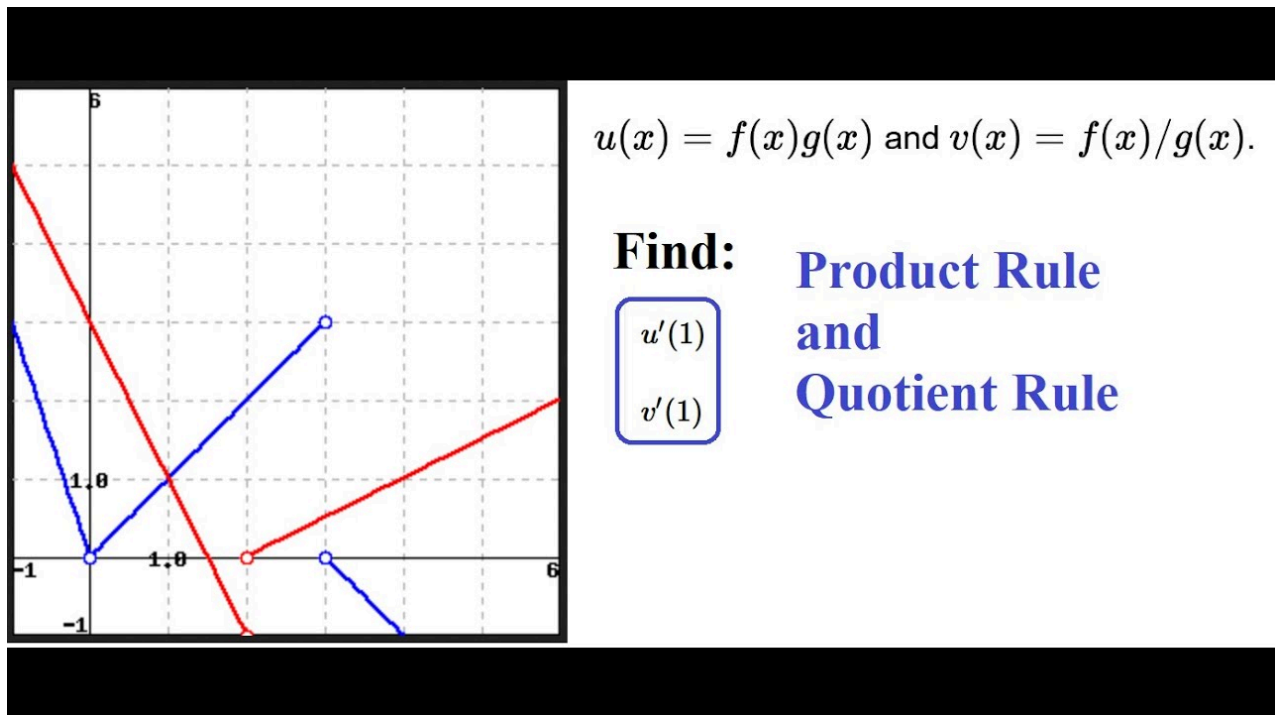
$$\frac{d}{dx} [f(g(x))] = f'(g(x)) \cdot g'(x)$$

In Leibniz Notation:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

Extended Chain Rule:

For multiple compositions: $\frac{dy}{dx} = \frac{dy}{dv} \cdot \frac{dv}{du} \cdot \frac{du}{dx}$



Graphical and formula-based illustration of the product rule and quotient rule for derivatives, showing functions and their derivatives.

Trigonometric Derivatives

Standard Trigonometric Functions

These formulas are fundamental and frequently tested in JEE:^{[1] [3]}

- $\frac{d}{dx} (\sin x) = \cos x$
- $\frac{d}{dx} (\cos x) = -\sin x$

- $\frac{d}{dx}(\tan x) = \sec^2 x$
- $\frac{d}{dx}(\cot x) = -\csc^2 x$
- $\frac{d}{dx}(\sec x) = \sec x \tan x$
- $\frac{d}{dx}(\csc x) = -\csc x \cot x$

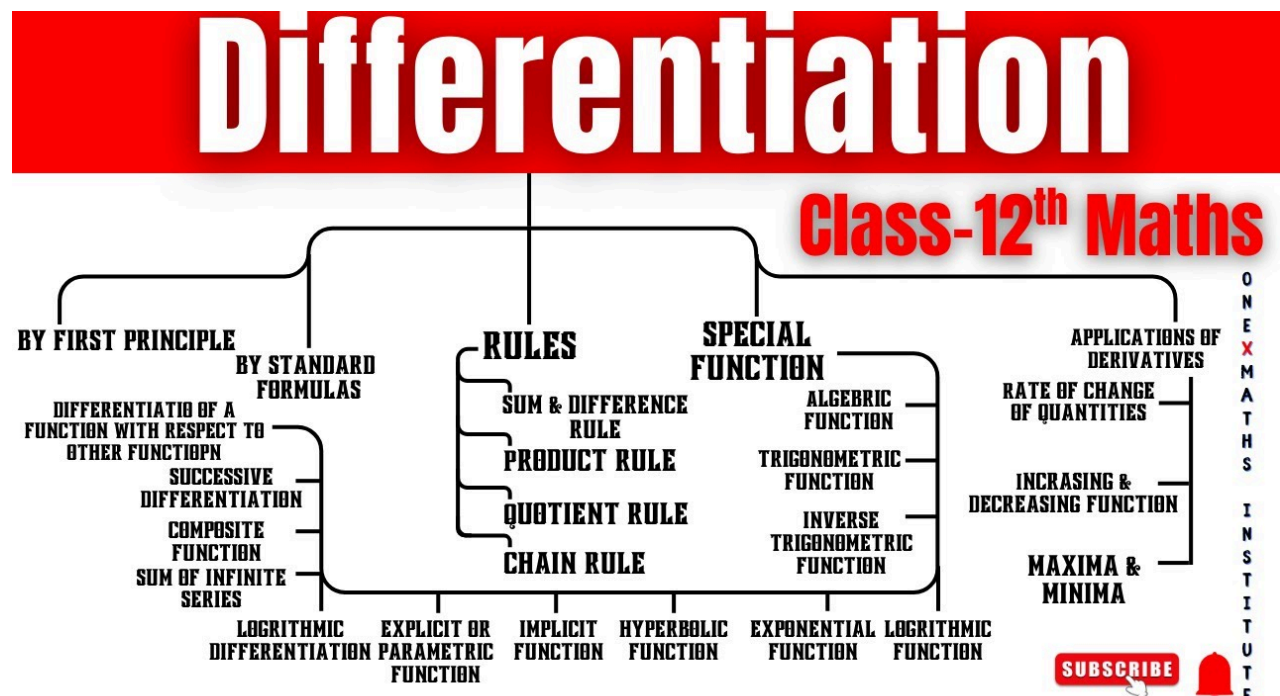
Trigonometric Identities in Differentiation

Important relationships:

- $\sec^2 x = 1 + \tan^2 x$
- $\csc^2 x = 1 + \cot^2 x$

Chain Rule Applications:

- $\frac{d}{dx}[\sin(ax + b)] = a \cos(ax + b)$
- $\frac{d}{dx}[\cos(ax + b)] = -a \sin(ax + b)$



Mind map summarizing differentiation topics and rules for Class 12 mathematics, including derivatives by first principle, standard formulas, special functions, and applications.

Exponential and Logarithmic Derivatives

Exponential Functions

Natural Exponential:

$$\frac{d}{dx}(e^x) = e^x$$

General Exponential:

$$\frac{d}{dx}(a^x) = a^x \ln a \quad (\text{where } a > 0, a \neq 1)$$

Chain Rule Applications:

- $\frac{d}{dx}(e^{f(x)}) = e^{f(x)} \cdot f'(x)$
- $\frac{d}{dx}(a^{f(x)}) = a^{f(x)} \ln a \cdot f'(x)$

Logarithmic Functions

Natural Logarithm:

$$\frac{d}{dx}(\ln x) = \frac{1}{x} \quad (x > 0)$$

General Logarithm:

$$\frac{d}{dx}(\log_a x) = \frac{1}{x \ln a} \quad (x > 0, a > 0, a \neq 1)$$

Chain Rule Applications:

- $\frac{d}{dx}[\ln(f(x))] = \frac{f'(x)}{f(x)}$
- $\frac{d}{dx}[\log_a(f(x))] = \frac{f'(x)}{f(x) \ln a}$

Inverse Trigonometric Derivatives

Standard Inverse Trigonometric Functions

These derivatives are crucial for JEE Advanced problems: ^[1] ^[8]

- $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}} \quad (|x| < 1)$
- $\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}} \quad (|x| < 1)$
- $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$
- $\frac{d}{dx}(\cot^{-1} x) = -\frac{1}{1+x^2}$
- $\frac{d}{dx}(\sec^{-1} x) = \frac{1}{|x|\sqrt{x^2-1}} \quad (|x| > 1)$
- $\frac{d}{dx}(\csc^{-1} x) = -\frac{1}{|x|\sqrt{x^2-1}} \quad (|x| > 1)$

Domain Restrictions

Critical Note: Always verify domain restrictions when using inverse trigonometric derivatives, as they have specific ranges where they are defined.

"INSTANTANEOUS RATE OF CHANGE"



Various ways
to denote a
Derivative

$$\frac{dy}{dx} \equiv \frac{d}{dx}(y) \equiv \frac{d}{dx}(f(x)) \equiv \frac{df}{dx} \equiv f'(x) \equiv f' \equiv y'$$

Algebra of Derivatives (RULES):

- $(u \pm v)' = u' \pm v'$
- $(uv)' = uv' + vu'$ (Product Rule)
- $\left(\frac{N}{D}\right)' = \frac{DN' - ND'}{D^2}$ (Quotient Rule) $D \neq 0$
- If $y = f(x) = u(v(w(x)))$ and so on, apply **CHAIN RULE**

$$\frac{dy}{dx} = \frac{du}{dv} \times \frac{dv}{dw} \times \frac{dw}{dx}$$

Direct way:

CHAIN RULE "Keep differentiating (and multiplying) all functions from the outermost to the innermost until you reach (X)!"

f(((((x))))))

Innermost Outermost

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Key differentiation rules including sum, product, quotient, and chain rules with formulas and annotations for class 11 and 12 mathematics and JEE preparation.

Advanced Differentiation Techniques

Logarithmic Differentiation

This technique is especially useful for complex expressions involving products, quotients, or powers: [6] [9]

Method:

1. Take natural logarithm of both sides
2. Differentiate implicitly
3. Solve for dy/dx

Example: For $y = x^x$

- $\ln y = x \ln x$
- $\frac{1}{y} \frac{dy}{dx} = \ln x + 1$
- $\frac{dy}{dx} = x^x (\ln x + 1)$

Implicit Differentiation

When y cannot be expressed explicitly as a function of x :^{[2] [10]}

Steps:

1. Differentiate both sides with respect to x
2. Apply chain rule where necessary (dy/dx terms)
3. Collect all dy/dx terms
4. Solve for dy/dx

Example: For $x^2 + y^2 = 1$

- $2x + 2y \frac{dy}{dx} = 0$
- $\frac{dy}{dx} = -\frac{x}{y}$

Parametric Differentiation

When both x and y are expressed in terms of a parameter t :^{[2] [10]}

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{y'(t)}{x'(t)}$$

Second Derivative:

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dt} \left(\frac{dy}{dx} \right) \cdot \frac{dt}{dx}$$

Higher Order Derivatives

Successive Differentiation

Notation:

- First derivative: $f'(x)$ or $\frac{dy}{dx}$
- Second derivative: $f''(x)$ or $\frac{d^2y}{dx^2}$
- n th derivative: $f^{(n)}(x)$ or $\frac{d^ny}{dx^n}$

Important Results:

- $\frac{d^nx}{dx^n}(e^{ax}) = a^n e^{ax}$
- $\frac{d^nx}{dx^n}(\sin x) = \sin\left(x + \frac{n\pi}{2}\right)$
- $\frac{d^nx}{dx^n}(\cos x) = \cos\left(x + \frac{n\pi}{2}\right)$

Leibnitz Theorem

For the nth derivative of a product uv:

$$(uv)^{(n)} = \sum_{r=0}^n \binom{n}{r} u^{(r)} v^{(n-r)}$$

Special Functions and Their Derivatives

Hyperbolic Functions

- $\frac{d}{dx}(\sinh x) = \cosh x$
- $\frac{d}{dx}(\cosh x) = \sinh x$
- $\frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$

Inverse Hyperbolic Functions

- $\frac{d}{dx}(\sinh^{-1} x) = \frac{1}{\sqrt{x^2+1}}$
- $\frac{d}{dx}(\cosh^{-1} x) = \frac{1}{\sqrt{x^2-1}} \quad (x > 1)$
- $\frac{d}{dx}(\tanh^{-1} x) = \frac{1}{1-x^2} \quad (|x| < 1)$

Problem-Solving Strategies

Choosing the Right Method

Decision Framework:

- Simple polynomials:** Use power rule directly
- Products:** Apply product rule
- Quotients:** Use quotient rule
- Composite functions:** Apply chain rule
- Complex expressions:** Consider logarithmic differentiation
- Implicit equations:** Use implicit differentiation
- Parametric forms:** Use parametric differentiation

Common JEE Problem Types

High-Frequency Topics:

- Chain rule with trigonometric and exponential functions
- Implicit differentiation of circles, ellipses, and curves
- Logarithmic differentiation for expressions with variable exponents
- Higher order derivatives and their applications
- Parametric differentiation and related rates

Error Prevention Tips:

1. Always check domain restrictions for inverse functions
2. Be careful with signs in trigonometric derivatives
3. Remember to apply chain rule for composite functions
4. Verify your answer by differentiating the result when possible

Applications in JEE Context

Rate of Change Problems

Differentiation provides tools to find:

- Velocity (derivative of position)
- Acceleration (second derivative of position)
- Marginal cost in economics
- Rate of change in physical quantities

Optimization Problems

Derivatives help in finding:

- Maximum and minimum values
- Critical points and inflection points
- Increasing and decreasing intervals
- Concavity and convexity

Equation of Tangents and Normals

Tangent at point (x_1, y_1) :

$$y - y_1 = f'(x_1)(x - x_1)$$

Normal at point (x_1, y_1) :

$$y - y_1 = -\frac{1}{f'(x_1)}(x - x_1)$$

Calculus

LIMITS & DERIVATIVES CHEAT SHEET

PROPERTIES OF LIMITS
$\lim_{x \rightarrow a} [cf(x)] = c \lim_{x \rightarrow a} f(x)$
$\lim_{x \rightarrow a} [f(x) \pm g(x)] = \lim_{x \rightarrow a} f(x) \pm \lim_{x \rightarrow a} g(x)$
$\lim_{x \rightarrow a} [f(x)g(x)] = \lim_{x \rightarrow a} f(x) \lim_{x \rightarrow a} g(x)$
$\lim_{x \rightarrow a} \left[\frac{f(x)}{g(x)} \right] = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}$ if $\lim_{x \rightarrow a} g(x) \neq 0$
$\lim_{x \rightarrow a} [f(x)]^n = \left[\lim_{x \rightarrow a} f(x) \right]^n$
LIMIT EVALUATIONS AT $\pm\infty$
$\lim_{x \rightarrow +\infty} e^x = \infty$ and $\lim_{x \rightarrow -\infty} e^x = 0$
$\lim_{x \rightarrow \infty} \ln x = \infty$ and $\lim_{x \rightarrow 0^+} \ln x = -\infty$
if $r > 0$: $\lim_{x \rightarrow \infty} \frac{c}{x^r} = 0$
if $r > 0$ & $\forall x > 0 [x^r \in \mathbb{R}]$: $\lim_{x \rightarrow -\infty} \frac{c}{x^r} = 0$
$\lim_{x \rightarrow \pm\infty} x^r = \infty$ for even r
$\lim_{x \rightarrow +\infty} x^r = \infty$ and $\lim_{x \rightarrow -\infty} x^r = -\infty$ for odd r
L'HOPITAL'S RULE
If $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{0}{0}$ or $\frac{\pm\infty}{\pm\infty}$ then $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$
DERIVATIVE DEFINITION
$\frac{d}{dx} [f(x)] = f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$
PRODUCT RULE
$[f(x)g(x)]' = f'(x)g(x) + f(x)g'(x)$
QUOTIENT RULE
$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$
CHAIN RULE
$\frac{d}{dx} [f(g(x))] = f'(g(x))g'(x)$
BASIC PROPERTIES OF DERIVATIVES
$[cf(x)]' = c[f'(x)]$
$[f(x) \pm g(x)]' = f'(x) \pm g'(x)$

COMMON DERIVATIVES	
$\frac{d}{dx}(x) = 1$	$\frac{d}{dx}[af(x)] = a \frac{d}{dx}[f(x)]$
$\frac{d}{dx}(ax) = a$	$\frac{d}{dx}(ax^n) = nax^{n-1}$
$\frac{d}{dx}(c) = 0$	$\frac{d}{dx}[f(x)]^n = n[f(x)]^{n-1}f'(x)$
$\frac{d}{dx} \left[\frac{1}{x^n} \right] = -nx^{-(n+1)} = -\frac{n}{x^{n+1}}$	
DERIVATIVES OF TRIGONOMETRIC FUNCTIONS	
$\frac{d}{dx}[\sin(x)] = \cos x$	$\frac{d}{dx}[\sec(x)] = \sec x \tan x$
$\frac{d}{dx}[\cos(x)] = -\sin x$	$\frac{d}{dx}[\csc(x)] = -\csc x \cot x$
$\frac{d}{dx}[\tan(x)] = \sec^2 x$	$\frac{d}{dx}[\cot(x)] = -\csc^2 x$
DERIVATIVES OF EXPONENTIAL & LOGARITHMIC FUNCTIONS	
$\frac{d}{dx}[e^x] = e^x$	$\frac{d}{dx}[a^x] = a^x \ln a$
$\frac{d}{dx}[\ln x] = \frac{1}{x}$	$\frac{d}{dx}[\ln x] = \frac{1}{x}, x > 0$
$\frac{d}{dx}[\log_a x] = \frac{1}{x \ln a}$	$\frac{d}{dx}[\ln f(x)] = \frac{f'(x)}{f(x)}$
$\frac{d}{dx}[e^{f(x)}] = f'(x)e^{f(x)}$	$\frac{d}{dx}[a^{f(x)}] = a^{f(x)} \ln a f'(x)$
$\frac{d}{dx}[f(x)^{g(x)}] = f(x)^{g(x)} \left(\frac{g(x)f'(x)}{f(x)} + \ln(f(x))g'(x) \right)$	
DERIVATIVES OF INVERSE TRIG FUNCTIONS	
$\frac{d}{dx}[\sin^{-1} x] = \frac{1}{\sqrt{1-x^2}}$	$\frac{d}{dx}[\sec^{-1} x] = \frac{1}{ x \sqrt{x^2-1}}$
$\frac{d}{dx}[\cos^{-1} x] = -\frac{1}{\sqrt{1-x^2}}$	$\frac{d}{dx}[\csc^{-1} x] = -\frac{1}{ x \sqrt{x^2-1}}$
$\frac{d}{dx}[\tan^{-1} x] = \frac{1}{1+x^2}$	$\frac{d}{dx}[\cot^{-1} x] = -\frac{1}{1+x^2}$
DERIVATIVES OF HYPERBOLIC FUNCTIONS	
$\frac{d}{dx}[\sinh x] = \cosh x$	$\frac{d}{dx}[\operatorname{sech} x] = -\operatorname{coth} x \operatorname{csch} x$
$\frac{d}{dx}[\cosh x] = \sinh x$	$\frac{d}{dx}[\operatorname{csch} x] = -\tanh x \operatorname{sech} x$
$\frac{d}{dx}[\tanh x] = 1 - \tanh^2 x$	
$\frac{d}{dx}[\operatorname{coth} x] = -1 - \operatorname{coth}^2 x$	

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Comprehensive calculus cheat sheet covering limits and derivative formulas essential for class 11th, 12th, and JEE preparation.

Conclusion

Mastering derivative formulas is essential for success in JEE Main and Advanced. The key lies in understanding when to apply each rule and technique rather than just memorizing formulas. Regular practice with different types of functions, combined with a solid grasp of the underlying principles, will build the confidence needed to tackle complex differentiation problems.

Remember that differentiation and integration are inverse operations, so understanding derivatives deeply will also strengthen your integration skills. Focus on recognizing patterns, understanding the geometric interpretation of derivatives as slopes of tangent lines, and applying these concepts to solve real-world problems.

The formulas and techniques presented here form the complete toolkit needed for JEE success. Practice systematically, starting with basic functions and progressively moving to more complex composite functions and advanced techniques like implicit and parametric differentiation.



1. <https://byjus.com/maths/differentiation-formulas/>
2. <https://www.vedantu.com/maths/differentiation-formula>
3. <https://www.jirs.ac.in/blogs/differentiation-formulas>
4. <https://allen.in/jee/maths/limits-and-derivatives>
5. <https://www.vedantu.com/question-answer/know-whether-to-use-chain-product-or-quotient-class-12-maths-cbse-6077b4f8a5e6f82a3752db7b>
6. <https://allen.in/jee/maths/methods-of-differentiation>
7. <https://byjus.com/maths/chain-rule/>
8. <https://www.orchidsinternationalschool.com/maths-concepts/differentiation-formulas>
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